2. $a=0 \quad b=1$

$$A. \begin{vmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 3 \end{vmatrix}$$

a. det A: $M_{11} = 1 \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} = 6-1 = 5$ $\quad M_{12} = 1 \begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix} = 3-1 = 2$ $\quad M_{13}\ 2 \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = 2(1-2 = -2$

$= 5-2-2 = 1$

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b. Matriks Minor

$M_{11} = 1\begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} = 5$ $\quad M_{12} = 1\begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix} = 2$ $\quad M_{13} = 2\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = -1$

$M_{21} = 1\begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} = 1$ $\quad M_{22} = 2\begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} = 1$ $\quad M_{23} = 1\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0$

$M_{31} = 1\begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = -3$ $\quad M_{32} = 1\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = -1$ $\quad M_{33} = 3\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} = 1$

$$A_M = \begin{vmatrix} 5 & 2 & -1 \\ 1 & 1 & 0 \\ -3 & -1 & 1 \end{vmatrix}$$

c. $\det(A) = 1$, maka $A^{-1} = Adj(A)$

Kof: $M_{11} = 5$ $\quad M_{12} = 2$ $\quad M_{13} = -1$

$M_{21} = 1$ $\quad M_{22} = 1$ $\quad M_{23} = 0$ $\quad \rightarrow (i+j)$ = genap = tetap

$M_{31} = -3$ $\quad M_{32} = -1$ $\quad M_{33} = 1$ $\quad (i+j)$ = ganjil = $\times(-1)$

$$C = \begin{vmatrix} 5 & -2 & -1 \\ -1 & 1 & 0 \\ -3 & 1 & 1 \end{vmatrix}$$

$Adj(A) = A^{-1}$ (jika det = 1)

$$Adj(A) = C^T = \begin{vmatrix} 5 & -1 & -3 \\ -2 & 1 & 1 \\ -1 & 0 & 1 \end{vmatrix}$$

E = N = 1, a = 0, b = 1

$$P = \begin{vmatrix} 120 & (50+N) & 200 \\ 80 & a+2 & (10+N) \\ (30+N) & 90 & 170 \end{vmatrix} \rightarrow \begin{vmatrix} 120 & 51 & 200 \\ 80 & 2 & 11 \\ 31 & 90 & 170 \end{vmatrix}$$

Enkripsi : $E = A.P$

Baris 1 : $1(120)+1(80)+2(31) = 262$, $1(51)+1(2)+2(90) = 233$, $1(200)+1(11)+2(170) = 551$

Baris 2 : $1(120)+2(80)+1(31) = 311$, $1(51)+2(2)+1(90) = 145$, $1(200)+2(11)+1(170) = 392$

Baris 3 : $1(120)+1(80)+3(31) = 293$, $1(51)+1(2)+3(90) = 323$, $1(200)+1(11)+3(170) = 721$

$$E = \begin{vmatrix} 262 & 233 & 551 \\ 311 & 145 & 392 \\ 293 & 323 & 721 \end{vmatrix}$$

→ Jika $\det(A) = 0$, maka tidak dapat dideskripsi, karena matriks enkripsi menjadi tidak punya invers. hal ini membuat data asli akan hilang.

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